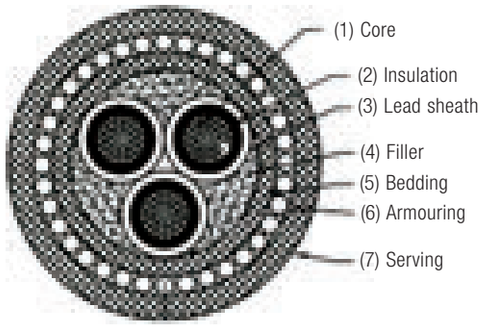


Fig. 1.19 S.L type cable

The structure of SL type cable is shown in figure 1.19 Even though this type is similar to that of H-type, each conductor is constructed with separate lead sleeves. Hence this cable can be handled easily for bending.

c) Pressure cables

A vacuum occurs when the solid power cord is used. It causes an electro chemical breakdown. Therefore, these pressure cables are used for distributing the voltage over 66 KV. The pressure of joint components used in these cable is increased and vacuum is reduced.

There are two types of pressure cables:

i. Oil filled cables
ii. Gas pump/pressure cables.

i. Oil filled cables

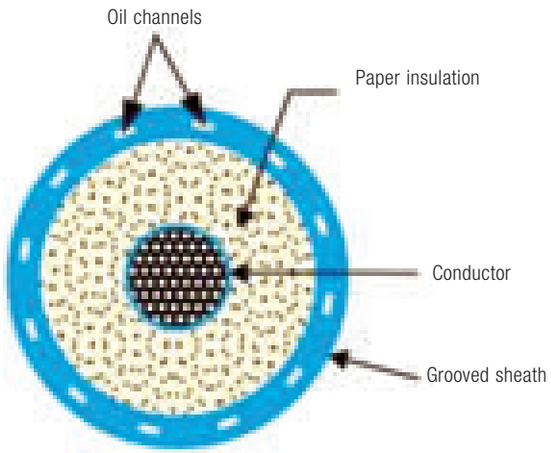


Fig. 1.20 Oil filled cables

The pipes are set up for oil circulation in oil-filled cables as shown in figure 1.20. The oil under pressure is kept supplied to the channel from a tank. The oil tank is set at a distance of about 500m along the cable route. Pressure oil is prevented from vacuum by pressing the paper insulation. This type of cable is used for distributing voltages from 66 KV to 230 KV.

ii. Gas filled cable

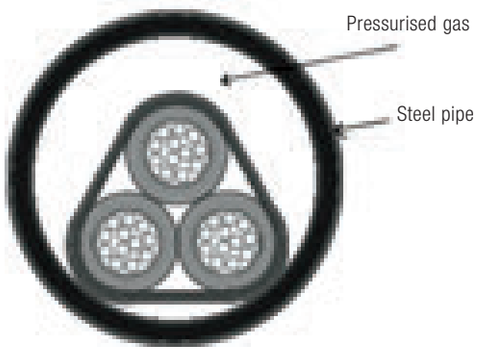


Fig. 1.21 Gas filled cable

Gas filled cable is shown in figure 1.21. The construction of gas pressure cable is similar to that of ordinary solid cable. It is designed in a triangular shape. The thickness of lead sheath of gas pressure cable is 75% of solid cable. The sheath is protected by a thin metal tape. The cable is kept in a gas tight steel pipe. The pipe is filled with dry nitrogen gas at 12 to 15 atmospheric pressure. The gas pressure produces radial compression and closes the voids formed between the layers of paper insulation. These cables carry more load current and operate at higher voltage than a normal cable. The nitrogen gas helps in quenching any flame. Its maintenance cost is low.

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